Depth Compression of GPT-2 Small by Adjacent-Layer Merging

A Unified Comparison of Learnable Weight-Space Merging, Iterative Operator Search, Affine Composition, and Similarity-Guided SLERP

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Code: https://github.com/sumitjhaa/gpt2-layermerging/tree/main/Assignment2

Abstract

We compressed GPT-2 Small (124,439,808 parameters across twelve transformer blocks) by folding adjacent blocks into single ones, and we judged the result by how much next-token perplexity we lost for each parameter saved. After the first presentation the mentor told us to try many different merging ideas, because adjacent-layer merging is barely explored; one rule held everywhere—only adjacent layers may be merged. Six of us covered the space: four main experiments, one tangential study that mapped the field and produced the baselines everyone else used, and a short pilot. Every experiment ran on the same WikiText-2 split for training and selection and the same three one-shot tests, DBpedia-14, AG News and IMDb, opened only at the very end.

Put simply, trained merges win. A learnable weight-space merger (Experiment A) beats plain deletion at every compression level and degrades gracefully instead of collapsing; a trainable block (Experiment B) is the best method in 14 of 15 unseen stage×dataset cells. Averaging does not work, because the neurons of two adjacent blocks are not aligned, and the cost at five merges runs from +1,385% (Experiment B) to +83,745% (Experiment A) in perplexity. Training-free rules are messier: affine composition with calibration-based selection (Experiment C) and calibrated SLERP (Experiment D) both beat deletion at one or two merges, but the winner changes with the pair and the model. Every per-pair ranking and control is in the body; the experiment we would run next is the same merging on a multilingual backbone.

1 Introduction

Decoder-only Transformer language models (Vaswani et al., 2017; Radford et al., 2019) store a large fraction of their parameters in *depth*. For GPT-2 Small each transformer block holds 5,898,240 parameters (4×768^2 attention projections, $2 \times 768 \times 3072$ MLP projections, biases and two LayerNorm vectors), so removing k of the 12 blocks removes $5.90k$M of 124.44M parameters. Depth compression is attractive because it reduces both memory and sequential latency, but naive removal of a block destroys the function that block computed.

We did this assignment as one team. After Presentation 1 the mentor told us that adjacent-layer merging is a brand-new area and that the group should tackle a *wide variety* of experiments, with each member pursuing one distinct idea; every method had to improve the perplexity–parameter-reduction trade-off, and non-adjacent layers must never be merged. Rather than have all six of us chase the same method, we agreed on a division of labour: each member designed and ran one distinct merging family under a shared protocol, and everyone's numbers are reported together here. The sections that follow are that merged write-up. Experiment A (Amrita) is a learnable *weight-space hypernetwork* that reads two adjacent blocks and emits dense additive corrections to the retained block, under a guarded deployment rule. Experiment B (Hiroyoshi) is an *iterative*, stage-wise operator search that compares deletion, averaging, closed-form affine composition and a trainable block under one exact evaluator. Experiment C (Lingala) is *affine* merging with a similarity-gated, error-weighted SLERP operator (SG-SC) and a calibration-guided operator selector (CGOS). Experiment D (Sravan) is a *calibrated*, similarity-guided SLERP merger whose per-module merge weights are tuned on a small calibration split. Sumit's foundational exploration is a tangential sweep of fifteen approaches that established the reference baselines—Block-Influence removal, pruned distillation, LayerDrop, a fitted ternary bridge—and the negative results that define the problem. Satvik's pilot covers the closed-form training-free merges (Union-Select-Reconstruct, Remove-and-Patch, Correlation-Guided Neuron Fusion).

Contributions.. We ran everything under one protocol—WikiText-2 for training and selection, with DBpedia-14, AG News and IMDb opened exactly once—so the four families can be compared on equal terms. The headline results, in order: a learned merger that beats deletion by up to 1.82 nats of validation cross-entropy at equal compression and reaches 68.23 IMDb perplexity at 28.48% removal (Experiment A); a controlled five-stage comparison in which a trainable block is best in 14 of 15 unseen stage×dataset cells (Experiment B); two training-free methods that beat deletion at low compression, affine composition with CGOS at 47.93 test perplexity (Experiment C) and a calibrated SLERP at 43.85 (Experiment D); and, at the level of how things go wrong, an explanation for why averaging fails (neuron permutation) and why local affine fitting does not automatically help—removability and residual low-rankness are different properties.

2 Report

2.1 Mentor Feedback Since Presentation 1

The study is organised around the feedback received after Presentation 1 and in the weekly mentor meetings.

- *Presentation 1 follow-up (group-wide).* The mentor asked us to "tackle a wide variety of experiments" because he sees adjacent-layer merging as a brand-new branch of research, subject to the hard constraint that *only adjacent layers may be merged*, and that the group must improve both perplexity and the parameter-reduction rate. We spread the work so that each member owns one distinct operator family (weight-space hypernetwork, iterative operator comparison, affine/CGOS, calibrated SLERP, and training-free closed-form merges) and the portfolio covers the design space.
- *Follow-up (multilingual).* The mentor noted that applying adjacent merges to *multilingual* models is interesting. That is why every operator here is backbone-agnostic, and why we name multilingual transfer as the primary follow-up (Section 3.9).
- *Week 1.* A learned merger must not be compared against a straw-man, and reconstruction loss alone is not evidence of language-modelling quality. We therefore let Experiment A start from DROP-FIRST exactly—its learned correction is zero-initialised—and added a temperature-scaled KL term; Experiment B promotes `Remove` to a first-class operator under one exact evaluator.
- *Week 2.* An approximate scorer could bias the comparison. Every experiment moved to exact, token-weighted sliding-window perplexity, and Experiment B added SHA-256 hashes to lock the protocol.
- *Week 3.* Demonstrate that merge semantics are correct before trusting results, and isolate selection from test. Experiment A gained a preflight equality check and a disjoint ranking partition; Experiment B runs five structural self-checks on a tiny GPT-2; Experiments C and D carve calibration/window holdouts and touch the test set once.
- *Week 4.* Hyperparameter ablation and explicit computational resources. Each experiment carries a targeted ablation (Tables 2 & 5 for A; 11 for B; SG-SC/rank sweeps for C; a weight-mode grid for D) and a resources paragraph (Section 2.5).
- *Week 5.* Keep the evaluation protocol identical across methods and state limitations honestly. We converged on one shared evaluator and committed to saying plainly where the evidence is thin; Section 3.9 does that.

2.2 Baselines and State of the Art

Depth reduction in Transformers has moved from simply deleting layers to learning how to fuse them, and we grouped the relevant work into four families, treating the closest members as explicit baselines.

Layer removal.. Gromov et al. (2024) show that the deeper layers of decoder-only Transformers are often redundant and can be deleted with little perplexity loss. Men et al. (2024) formalise this with the *Block Influence* (BI) score, the cosine similarity between a block's input and output, and remove the least influential blocks. The DROP-FIRST control in Experiments A and B, the `Remove` operator in Experiments B and C, and the foundational BI baseline are exactly this family.

Static layer merging.. Yang et al. (2024) collapse rear layers into preceding ones with the Reserving-Differences-while-Seeking-Common (RDSC) rule and no training. Liu et al. (2024b) identify mergeable layers by manifold alignment in activation space and combine them by closed-form averaging. Both use *fixed* coefficients from a similarity heuristic. The `Average` operator (Experiments B, C, D) and the Random-Weight-Average control (Experiment C) represent this family.

Affine composition.. `LinearMultiply` (Experiment B) fits a linear map from a group's input to its output and absorbs the span into a single affine block, composed layer-wise through Cholesky factors. It is the strongest *training-free* operator in the study and is related in spirit to the closed-form composition used by Yang et al. (2024).

Learned fusion.. Pei et al. (2024) introduce FuseGPT, which reframes pruning as knowledge redistribution: a redundant block is fused into a neighbour using low-rank learnable matrices under a Macro Influence criterion with lightweight local distillation. This is the closest SOTA and the upper-reference target. Our learnable methods follow the same philosophy—the merged artifact is *trained* to reproduce the hidden states of the original span plus a logit-level KL term (Hinton et al., 2015)—but differ in form: Experiment A uses a weight-space hypernetwork with *dense* corrections, and Experiment B uses a full trainable GPT-2 block.

Distillation and structured pruning.. Sanh et al. (2019) halve BERT's depth by distillation with every-other-layer initialisation. Orthogonal structured methods delete rows/columns (SliceGPT, Ashkboos et al. 2024) or non-critical channels (LLM-Pruner, Ma et al. 2023); they do not reduce depth and serve as context, although our distillation repairs follow the LLM-Pruner recipe. Fan et al. (2020) fine-tune with structured layer dropout; we ran it as a full competing method in the foundational exploration because it is the one depth-reduction scheme with a public recipe. Stoica et al. (2024) merge features within and across models, the closest relative of our fusion pilot. The similarity-guided pair selection and adaptive merge weights of Experiments C and D are motivated by Zhang et al. (2025) and Liu et al. (2024a), and the ternary factor grid by Ma et al. (2024).

Availability of baseline implementations.. The removal and averaging controls are implemented by us in full, so they need no external release. The same is true of `LinearMultiply`, the four operators of Experiment B, the SG-SC and CGOS selectors of Experiment C, the calibrated SLERP of Experiment D, and the fifteen foundational approaches; everything is directly comparable by construction. The closest external method, FuseGPT, has no official public release, and its low-rank fusion is tied to its own pruning schedule. We therefore position against its *idea* rather than a re-run, instantiating every reproducible mechanism as a like-for-like control.

2.3 Datasets

Training and selection data.. All learning, calibration and model selection use only **WikiText-2** (Merity et al., 2017), loaded as `Salesforce/wikitext`, config `wikitext-2-raw-v1`. Raw rows are reassembled into whole articles where needed, exact duplicate articles are removed, and the unique text is partitioned *before* tokenisation into disjoint train/checkpoint/ranking (A) or train/calibration/evaluation (C, D) roles with a fixed permutation and seed; every partition and its hash are recorded.

In-domain final evaluation.. The full official WikiText-2 validation split is used only at the end and only for the selected configuration; the scored-token counts range from 244,824 (A) to 249,748 (B) across experiments because of small differences in boundary-token treatment. The original model reaches PPL = 29.37 (A), 28.61 (B), and 33.94 (D, on its own validation windows); we never compare absolute values across experiments because the exact evaluators differ in window size and stride.

Out-of-domain test data.. We evaluate on three external test sets chosen so that their relationship to WikiText-2 varies along two axes, *context* and *tokens*, exactly as the assignment directs: *DBpedia-14* (Zhang et al., 2015; Lehmann et al., 2015) (similar context, different tokens), *AG News* (Zhang et al., 2015) (different context, nearer tokens), and *IMDb* (Maas et al., 2011) (different in both). Each source is capped at a fixed token budget with a fixed seed (32,768 tokens in A; 400,000 in B); labels are ignored and only next-token prediction is scored. Because the corpora differ in content and token cap, absolute perplexities are *not* comparable across domains; we report relative change over each corpus's own original GPT-2 baseline.

Bias handling.. WikiText-2 is a single-domain English Wikipedia corpus, so selection can quietly inherit its lexical and topical biases. We tried to keep that from contaminating the comparison. Articles are partitioned before tokenisation at article granularity, so near-duplicates cannot leak across roles, and exact duplicates are removed before the split. Model selection never touches the official validation split or the external corpora—A ranks on a disjoint holdout under a guarded acceptance rule, B selects on validation and locks the test text with hashes, and C and D select on calibration or holdout windows. Finally, unseen domains are reported relative to their own GPT-2 baseline, so a vocabulary mismatch between corpora cannot dress itself up as a compression effect.

Bottleneck.. The dominant bottleneck varies by experiment. In A it is GPU memory during logit materialisation, handled by scoring in chunks of 128 tokens under activation checkpointing; the secondary bottleneck is the exhaustive

non-overlapping-set search for k merges. In B it is evaluation and distillation cost, so each unseen corpus is capped and the evaluator batched; the second bottleneck is *stage-wise coupling*. In C and D it is the calibration/activation pass and the affine solve. The foundational exploration's bottleneck is that all fifteen approaches ran on one free Apple-Silicon MPS laptop with no paid GPU, forcing 60–80-step training budgets. The common bottleneck is two shared six-hour-class accelerators, which caps training passes.

Dataset extension and difference from the previous version.. Relative to Presentation 1, which used a single WikiText-2 split, the data pipeline changed in three ways. Articles are now re-partitioned into disjoint roles with a fixed seed and hashes, so the selection holdouts cannot overlap the training text. Three external corpora (DBpedia-14, AG News, IMDb) were added as a one-shot unseen test along the two axes the assignment points at. And small calibration budgets (~4k–20k tokens) are carved out of the *train* split, so no fitted operator ever sees validation or test text. The old setup could only measure in-domain fit; the new one measures held-out selection and out-of-domain transfer separately. No new data were collected; all corpora are public.

2.4 Experimental Plan

Research questions.. We went in with four questions. First, whether a *learned* artifact could fold one block into its neighbour and still beat plain deletion on held-out text. Second, which adjacent pairs are actually safe to merge, and whether the ranking a model learns at home still holds once we move to out-of-domain data. Third, how quality falls off as we keep merging, and whether the ordering between operators survives into the unseen domains. Fourth, how close a training-free rule can get to the learned methods, and whether a cheap calibration signal agrees with validation-based selection.

Setting and robustness under limited compute.. Everything ran on shared six-hour Kaggle/Colab sessions with two NVIDIA Tesla T4 GPUs (15 GiB each), 4 vCPU cores, ~13 GiB RAM and ~20 GiB scratch, plus the Apple-Silicon laptop for the foundational grid. Within that budget we stayed robust by mixing training tasks so that every pair and compression level appears in a single pass (A), by capping the external corpora and batching the evaluator rather than shortening training (B), by preferring closed-form fits wherever one exists (B, C, D and the pilot), and by deriving every stochastic fit from an explicit per-fit generator whose seeds are recorded (foundation).

How the approaches differ.. The four primary experiments are separated by *mechanism*, not by incidental implementation. What gets learned or selected changes across them: a dense additive correction in *weight space* (A), a full replacement block in *activation space* (B), a closed-form affine map *picked* by calibration (C), or a per-module *interpolation weight* on a spherical path (D). The search differs too, from exhaustive non-overlapping-set selection (A) and a greedy stage-wise boundary search (B) to per-pair then joint selection (C) and greedy similarity-guided iteration (D). So does the compute: gradient training of a hypernetwork and of a single block in A and B, versus the training-free closed-form and interpolation operators of C, D and the pilot. The dependent variable stays the same everywhere—next-token cross-entropy, reported as perplexity, with top-1/top-5 accuracy and parameter count as secondary measures.

Model architectures and parameter counts.. All experiments operate on GPT-2 Small: $n_{\text{layer}} = 12$, $d = 768$, 12 heads, vocabulary 50,257, tied input/output embedding, learned positional embeddings, and `Conv1D` projections stored as [in, out]; the total is 124,439,808 parameters and each block holds 5,898,240. Merged artifacts differ in size: A's training-time hypernetwork holds 39,391,168 parameters (discarded at deployment); B's `NonlinearMultiply` holds one block's parameters; C's affine map holds $768^2 + 768$ parameters while dropping the rest of the block; D's SLERP added no new parameters in the reported runs; the pilot's low-rank patch holds $\sim 2dr$ parameters; and the foundational ternary bridge holds $\sim 2dr$ weights at ~1.58 bit/param. Deployed compressed models after k merges hold $124{,}439{,}808 - 5{,}898{,}240\,k$ parameters (viz. 117,351,936 at $k = 1$ and 89,000,448 at $k = 5$).

Hyperparameter search and ablation.. A full grid is infeasible on the shared GPUs, so we run targeted ablations and state explicitly what is fixed: *(A)* an exact throughput-and-memory sweep over global batch size (Table 2) and a training-length sweep reading the equal-level CE gap per epoch (Table 5), all other weights fixed (Table 3); *(B)* a targeted ablation on the nonlinear training length (Table 11); *(C)* SG-SC component ablations, a rank/direction sweep, and CGOS vs. validation selection; *(D)* a merge-weight/mode grid (fixed-0.5 linear, similarity-prior SLERP, calibrated shared weight, calibrated per-group weights with mode search); and the *foundation* a rank/bitwidth sweep with a random-removal control.

Metric choice and justification.. The primary metric is mean next-token cross-entropy and its exponential, perplexity. Cross-entropy is the right scalar because the task is pure language modelling: it is additive over positions and strictly proper, so a model cannot be rewarded for a confidently wrong distribution. Perplexity is reported alongside as the conventional unit in depth-pruning work and is used only for *within-corpus* comparisons. Beyond that we also kept an eye on top-1 and top-5 next-token accuracy over the same sliding windows, the paired cross-entropy gap against the matching control, per-window NLL variance, an internal hidden-state diagnostic (the normalised MSE between the merged block's output and the teacher span), parameter count, fp16 bytes, single-sequence latency, and the calibration–validation selection agreement in C. We deliberately skipped BLEU and ROUGE; this is a preservation study, not a generation study, so string-repetition metrics would have told us nothing.

Criteria.. A gain is reported only when (i) it is measured on held-out text, (ii) it beats the matching control, and (iii) it survives the experiment's selection/guard rule. Unseen-domain data are used only for final reporting, never for selection.

System effort and demonstration.. Most of the system effort went into four awkward corners: the weight-tokenisation/unpack round-trip and the two-GPU synchronous schedule (A), the stage-wise plan search and hash-locked exact evaluator (B), the calibration plumbing with a Cholesky affine solve (C), and the greedy similarity-guided iteration with calibration tuning (D). We also kept a single narrative that runs the whole portfolio top to bottom: load GPT-2 and print its 124,439,808 parameters; run the structural self-checks that certify merge semantics before any result is trusted; show that the untrained Experiment A merger is bit-identical to DROP-FIRST and that its ranking CE gap only turns negative once training starts; display the Experiment B unseen matrix and the transition from training-free to learnable operators; let CGOS reproduce the validation-best operator on 9 of 11 pairs while the calibrated SLERP beats fixed-0.5 averaging; and finally open the locked unseen test set exactly once to reproduce the headline numbers.

2.5 Project Management

How the novel solutions were proposed.. The portfolio was organised *around a design-space map*, not around a single idea. The foundational exploration first established that deletion is the strict control and that weight averaging fails because of neuron-permutation mismatch (Section 3.6); this made the central question falsifiable ("does a learned artifact beat deletion at equal parameter count?"). Each member then proposed one mechanism at a different point on the learned/training-free and weight-space/activation-space axes.

Computational resources.. Two NVIDIA Tesla T4 GPUs (15 GiB each), 4 vCPU cores, ~13 GiB RAM, ~20 GiB scratch, plus one Apple-Silicon laptop (MPS, 16 GiB, no paid GPU) for the seeded studies and the foundational grid. Peak per-GPU memory during A's batch probe was under 1.4 GiB, so A is GPU-compute-bound rather than memory-bound; its ten training passes took 4174 s and the full pipeline fits in one session. A's 39.4M trainable parameters are the largest memory item but are discarded at deployment.

Task division.. Each member owned one distinct operator family and its notebook, while dataset curation, the shared evaluation protocol, the baseline/SOTA framing and cross-checking were shared. The full allocation is in Section 4.

3 Experiment

3.1 Shared Protocol and Preliminaries

GPT-2 Small has $n_{\text{layer}} = 12$, $d = 768$, 12 attention heads, vocabulary 50,257, a tied input embedding W_e and output head, and learned positional embeddings W_p. For an input id sequence x the initial hidden state is $s_0 = W_e[x] + W_p[0{:}T]$, and each pre-LayerNorm block computes

$$u_i = s_i + \text{MHA}_i(\text{LN}(s_i)), \qquad s_{i+1} = u_i + \text{MLP}_i(\text{LN}(u_i)), \tag{1}$$

for $i = 0, \ldots, 11$, with a final layer norm LN_f before the LM head. Perplexity is computed by an exact, token-weighted sliding-window evaluator with context 512 and stride 256: for each window we score the newly covered labels, weight the negative log-likelihood by the number of new tokens, and aggregate

$$\text{PPL} = \exp\left(\frac{\sum_w n_w \, \text{NLL}_w}{\sum_w n_w}\right), \tag{2}$$

so every token is counted exactly once. Throughout, a *plan* is a tuple of contiguous layer groups $G = (g_1, \ldots, g_m)$ partitioning $\{0, \ldots, 11\}$, and a compression step merges two *adjacent* current groups at a boundary b, replacing

Tensor	Kind	Shape	Patch	Tokens
q_w, k_w, v_w, o_w	square	768^2	64^2	144 each
up_w	up	768×3072	64×256	144
$down_w$	down	3072×768	256×64	144
q_b, k_b, v_b, o_b	vector	768	64	12 each
up_b	vector	3072	64	48
$down_b$	vector	768	64	12
$\ln_1, \ln_2$ w/b	vector	768	64	12 each
Total per block				1020
Total per pair				2040

Table 1: Experiment A parameter-tensor inventory and patch tokenisation.

(g_b, g_{b+1}) by $g_b \cup g_{b+1}$; we write $[i, j]$ for a group spanning original blocks $i, \ldots, j$.

3.2 Experiment A: Second-Block-Anchored Learnable Weight Merging

Idea.. A compact bidirectional Transformer (a *hypernetwork over weights*) reads the parameter tensors of two adjacent blocks and emits *additive corrections* to the parameters of the *second* block so that it absorbs the function of the first. Because every correction head is zero-initialised, the untrained model is *exactly* DROP-FIRST and the deployed architecture is byte-identical to GPT-2; this gives a clean starting point and a strict control.

Weight tokenisation into patch tokens.. Every parameter tensor of a block is flattened into fixed-size patches that become tokens for the merger. A vector of length 768 is split into 12 chunks of 64; a vector of length 3072 into 48 chunks. A $(768, 768)$ matrix is tiled by 64×64 patches (144 patches); the MLP up-projection $(768, 3072)$ uses 64×256 patches and the down-projection $(3072, 768)$ uses 256×64 patches (144 each):

$$\text{pack}_{\text{vec}}(x) = x.\text{reshape}(L/64,\ 64), \tag{3}$$

and for $X \in \mathbb{R}^{H \times W}$ with patch (p_h, p_w),

$$\text{pack}_{\text{mat}}(X) = X.\text{reshape}\Big(\tfrac{H}{p_h}, p_h, \tfrac{W}{p_w}, p_w\Big).\text{permute}(0, 2, 1, 3).\text{reshape}(-1,\ p_h p_w)\,, \tag{4}$$

with an inverse `unpack`. Each block yields $6 \times 144 + (4 \times 12 + 48 + 12 + 4 \times 12) = 1020$ tokens; a pair yields 2040 tokens (Table 1).

Architecture.. For a pair (A, B) and left index ℓ, each family f with kind k is rescaled by a per-family scale $\sigma_f = \max(\sqrt{\tfrac{1}{2}(\overline{A_f^2} + \overline{B_f^2})},\ 0.01)$, where $\bar{\cdot}$ is the mean over tensor elements. Each source $r \in \{A, B\}$ receives a token sequence

$$\begin{aligned} t_p^{f,r} = \text{Enc}_k\Big(\tfrac{\text{pack}(r_f)_p}{\sigma_f}\Big) + e_{\text{row}}(p_r) + e_{\text{col}}(p_c) \\ + e_{\text{fam}}(f) + e_{\text{src}}(r) + e_{\text{layer}}(\ell + [r = B]), \end{aligned} \tag{5}$$

with learned embeddings $e_{\text{fam}} \in \mathbb{R}^{16 \times d_m}$, $e_{\text{src}} \in \mathbb{R}^{2 \times d_m}$, $e_{\text{layer}} \in \mathbb{R}^{12 \times d_m}$, $e_{\text{row}}, e_{\text{col}} \in \mathbb{R}^{48 \times d_m}$ and a kind-specific linear encoder. All 2040 tokens pass through six unmasked (bidirectional) pre-LN blocks of width $d_m = 384$ and 6 heads,

$$a = \text{softmax}\Big(\tfrac{QK^\top}{\sqrt{d_h}}\Big) V, \tag{6}$$

$$x \leftarrow x + W_o\, a, \tag{7}$$

$$x \leftarrow x + W_2\, \text{GELU}(W_1 \text{LN}_2(x)), \tag{8}$$

with no causal mask. Writing $z = [z_A; z_B]$, the aligned halves are fused position-wise,

$$m_p = W_f\, \text{GELU}\big(\text{LN}([z_{A,p}; z_{B,p}])\big)\,, \tag{9}$$

and a kind-specific decoder emits the correction for the retained block,

$$\Delta_f = \sigma_f \cdot \text{Unpack}_k(\text{Dec}_k(m))\,, \qquad \hat{B}_f = B_f + \Delta_f. \tag{10}$$

The zero-initialised Dec_k make $\Delta_f \equiv 0$ at initialisation, so the untrained merger is exactly DROP-FIRST; the merger holds 39,391,168 parameters. The merger itself is never deployed: after training only the corrected second-block tensors are exported, so the client always receives a standard GPT-2, and at $k = 5$ a model with 89,000,448 parameters (a saving of 35,439,360).

Global batch	Step (s)	Tokens/s	GPU peak (GiB)
2	0.720	710.7	1.20
4	0.745	1375.2	1.23
8	0.819	2500.7	1.26
16	1.028	3983.1	1.34

Table 2: Experiment A batch auto-tuning probe (backward passes, no optimiser updates); the maximum of the two GPUs is shown. Batch 16 is selected.

Hyperparameter	Values	Basis	Selected
Batch size	2, 4, 8, 16	tokens/s, memory	16
Training passes	$0, \ldots, 10$	CE gap	10
LR (AdamW)	10^{-4} (fixed)	prior work	10^{-4}
Weight decay	0.01 (fixed)	prior work	0.01
Grad. clip	1.0 (fixed)	stability	1.0
Temperature T	2.0 (fixed)	prior work	2.0
λ_{KD}	1.0 (fixed)	balancing	1.0
λ_{corr}	0.01 (fixed)	scale prior	0.01
λ_{CE}	0.0 (fixed)	preservation	0.0
Guard margin	0.001 (fixed)	conservative	0.001

Table 3: Experiment A hyperparameter summary; only batch size and training length are measured ablations.

Training objective.. With teacher states $s_0, \ldots, s_{12}$ and student state h, for a merged-pair set $\mathcal{P}$ the hidden objective averages a normalised MSE over the replaced blocks,

$$\mathcal{L}_{\text{hid}} = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \frac{\left\| h^{(i+2)} - s_{i+2} \right\|_F^2}{\overline{(s_{i+2})^2} + 10^{-8}}, \tag{11}$$

where $h^{(i+2)}$ is the student state after replacing pair $(i, i+1)$, so later merges see earlier ones. With temperature $T = 2$ the distillation term is $\mathcal{L}_{\text{KD}} = T^2 \cdot \frac{1}{N} \sum_n \text{KL}(p_n^{\text{tea}} \| p_n^{\text{stu}})$, and a relative correction penalty discourages large departures from B,

$$\mathcal{L}_{\text{corr}} = \frac{1}{16} \sum_f \frac{\overline{(\hat{B}_f - B_f)^2}}{\max\left(\frac{1}{2}(\overline{A_f^2} + \overline{B_f^2}),\ 10^{-4}\right)}, \qquad \mathcal{L} = \mathcal{L}_{\text{hid}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}} + \lambda_{\text{corr}} \mathcal{L}_{\text{corr}} + \lambda_{\text{CE}} \mathcal{L}_{\text{CE}}, \tag{12}$$

with $\lambda_{\text{KD}} = 1.0$, $\lambda_{\text{corr}} = 0.01$, $T = 2.0$ and $\lambda_{\text{CE}} = 0$ (ground-truth CE is logged but receives no weight, making this a pure preservation study). Logits and CE are computed in chunks of 128 tokens with gradient checkpointing.

Optimisation and two-GPU schedule.. AdamW at LR 10^{-4}, weight decay 0.01, gradient clipping 1.0, FP16 autocast. Two frozen GPT-2 copies (one per GPU); before each step the merger parameters are copied to the secondary device as one contiguous transfer; the batch is split into two shards whose losses are weighted and summed, and gradients from the secondary are added—exact synchronous data parallelism. A batch auto-tuner selects the fastest global batch below 85% memory. Ten epochs traverse the retained training partition once each (21,273,600 token presentations over 2,127,360 unique tokens) in 4174 s.

Local-loss ablation.. A completed ablation holds the six-block merger architecture, the vocabulary-KL term and the correction penalty fixed and changes only the local loss. Hidden MSE alone gives the lowest raw learned PPL on 10 of 11 pairs; adding a feature-KL term improves the L3–L4 pair slightly but ruins L1–L2 (Table 4), where the first block's embedding-level transform is poorly matched in feature space. Hidden MSE is therefore retained in the main method.

Selection, guarding and controls.. Single pairs are ranked by their guarded perplexity change on the ranking hold-out,

$$\Delta\text{PPL}_i = 100 \left(\exp\left(\text{CE}_i^{\text{learned}} - \text{CE}^{\text{orig}} \right) - 1 \right), \tag{13}$$

and a pair is accepted only if $\text{CE}^{\text{learned}} \le \text{CE}^{\text{drop-first}} - 0.001$. For a fixed k the plan is

$$\mathcal{P}_k^\star = \underset{|P|=k,\ p_{j+1} > p_j + 1}{\arg\min} \sum_{i \in P} \Delta\text{PPL}_i, \tag{14}$$

Local loss	L1–L2 PPL	L2–L3 PPL	L3–L4 PPL
Hidden MSE only	74.35	33.22	39.08
Feature KL only	4550.61	33.48	35.16
MSE + feature KL	3872.49	33.44	35.16

Table 4: Experiment A local-loss ablation over three representative pairs with the same architecture, vocabulary KL and correction penalty; each variant ran ten epochs. MSE is used as the local loss in the reported results.

Epoch	CE gap vs DROP-FIRST	New best
0	0.00000	0.00000
1	−0.42714	−0.42714
2	−1.00012	−1.00012
3	−1.39179	−1.39179
4	−1.54557	−1.54557
5	−1.68116	−1.68116
6	−1.70291	−1.70291
7	−1.75209	−1.75209
8	−1.80910	−1.80910
9	−1.75780	−1.80910
10	−1.81972	−1.81972

Table 5: Experiment A equal-level validation CE gap versus DROP-FIRST by epoch. Negative is better; the selected checkpoint is epoch 10.

computed by exhaustive enumeration of non-overlapping edges; joints are then measured independently because this minimises an additive surrogate, not the true joint perplexity. Controls on the same pairs are DROP-FIRST, DROP-SECOND, and arithmetic averaging $\hat{b} = (a + b)/2$.

Results: checkpoint selection.. The validation CE gap versus DROP-FIRST improves monotonically through epoch 8 and reaches −1.82 nats at epoch 10 (Table 5); the initial drop-first checkpoint (epoch 0, gap 0.0) is included so the improvement cannot be attributed to initialisation.

Results: single-pair ranking.. Table 6 reports all 11 adjacent pairs, ranked by guarded perplexity change on the 16,384-token ranking holdout. The safest pair is $L2$-$L3$ (+2.59%); the next eight pairs lie between +6.71% and +13.19%; $L3$-$L4$ is rejected by the guard; and $L1$-$L2$ is catastrophic for deletion (PPL = 6763.70), showing that the first block applies an essentially irreplaceable embedding-level transformation.

Results: joint plans.. Table 7 shows the selected joint plans for $k = 1 \ldots 5$. The minimum-additive-cost rule may repeat a pair across k, so the selections need not nest; every selected joint beats its matching DROP-FIRST control, and the ranking CE gap grows from −0.025 to −0.805 nats.

Results: in-domain and unseen-domain.. On the official WikiText-2 validation split a single merge costs 1.02 perplexity (+3.48%) while removing 5.70% of parameters, and top-1 accuracy falls only from 0.3826 to 0.3763 (Table 8). One merge costs almost nothing on the unseen sets either (+0.06% on DBpedia-14, +6.64% on AG News, +3.22% on IMDb). Table 9 gives the full picture for $k = 0 \ldots 5$ and Figure 1 plots it: perplexity grows smoothly with depth instead of collapsing, and at $k = 5$ (28.48% removed) the learned model reaches 47.31 (WikiText-2), 70.41 (DBpedia-14), 139.15 (AG News) and 68.23 (IMDb), against DROP-FIRST controls of 98.41, 106.72, 173.99 and 90.52. Top-1 and top-5 accuracy at $k = 5$ fall by 4–10 points depending on the dataset (Table 10). Across the twenty dataset×compression cells the learned model beats DROP-FIRST in eighteen of them (Fig. 2 shows the WikiText-2 analogue); the only exceptions are AG News at one and two merges, where deletion is ahead by 0.24 and 1.17 perplexity, a range that is close to the evaluation noise of the method itself. Following the run artefacts, all numbers here are the *raw* learned model; a guarded variant falls back to DROP-FIRST per pair when the ranking margin fails, but that guard is not binding for any selected plan.

Preflight checks.. Asserted and recorded before training: (i) `pack`/`unpack` round-trips for every kind; (ii) `teacher_states`$\circ \mathrm{LN}_f$ equals the model's own `last_hidden_state`; (iii) every generated pair equals DROP-FIRST exactly at initialisation (rtol = 0); (iv) a joint learned plan equals the matching DROP-FIRST plan at initialisation; (v) the KL of identical models is zero; and (vi) every generated second-block tensor receives a finite non-zero gradient

Rank	Pair	Guard	Learned	Drop-first	ΔPPL
1	L2-L3	acc.	33.21	34.04	+2.59%
2	L4-L5	acc.	34.54	37.16	+6.71%
3	L5-L6	acc.	34.93	39.16	+7.89%
4	L9-L10	acc.	35.45	39.10	+9.52%
5	L7-L8	acc.	35.68	38.26	+10.22%
6	L6-L7	acc.	35.81	39.66	+10.62%
7	L8-L9	acc.	35.84	38.99	+10.71%
8	L11-L12	acc.	36.01	43.30	+11.26%
9	L10-L11	acc.	36.64	39.57	+13.19%
10	L3-L4	rej.	39.13	36.95	+14.14%
11	L1-L2	acc.	73.69	6763.70	+127.64%

Table 6: Experiment A single-pair ranking on the WikiText-2 ranking holdout. ΔPPL is the learned model's percentage increase over the *original uncompressed* model (Eq. 13), *not* relative to DROP-FIRST.

k	Merged pairs	Additive cost	Removed	CE gap vs DROP-FIRST
1	L2-L3	2.589	5.70%	−0.025
2	L2-L3; L4-L5	9.304	11.39%	−0.160
3	L2-L3; L4-L5; L9-L10	18.827	17.09%	−0.278
4	L2-L3; L4-L5; L7-L8; L9-L10	29.048	22.78%	−0.468
5	L2-L3; L4-L5; L7-L8; L9-L10; L11-L12	40.303	28.48%	−0.805

Table 7: Experiment A selected joint plans and their ranking-holdout CE gap versus the matching DROP-FIRST control. Negative is better.

while all GPT-2 parameters receive none.

System effort and demonstration assets.. The system covers dataset preparation, weight patching, differentiable weight generation, dual-GPU optimisation, model reconstruction and evaluation. A live demo can load a saved pair replacement, re-run the frozen-teacher checks, and compare the original model, the drop baselines, parameter averaging and the learned weights on the same tokens; the saved tables and plots show the one-to-five-merge trade-off without any retraining during the demonstration.

3.3 Experiment B: Multi-Step Operator Comparison

Idea.. Treat compression as a sequence of five steps. At every step we enumerate every boundary between currently adjacent groups, apply one of four operators, and keep the candidate that minimises exact WikiText-2 validation NLL. The only thing that changes between runs is the operator itself, which is what makes the comparison clean.

Operators.. A plan is a tuple of contiguous groups $G = (g_1, \ldots, g_m)$; a step merges (g_b, g_{b+1}) into $g_b \cup g_{b+1}$.

- `Remove`: group $[i, j]$ is replaced by its first block, $\hat{B} = B_i$.
- `Average`: every tensor is averaged over the group's *original* blocks, $\hat{\theta} = \frac{1}{j-i+1} \sum_{\ell=i}^{j} \theta_\ell$.
- `LinearMultiply` (affine composition): for a span $[i, j]$ collect input/output activations (X_ℓ, Y_ℓ) at each boundary and fit, per consecutive pair, a ridge-regularised affine map

$$A^\star, b^\star = \arg\min_{A,b} \|X_c A - Y_c\|_F^2 + \lambda \|A\|_F^2, \quad \lambda = \rho \, \frac{\mathrm{tr}(X_c^\top X_c)}{d}, \tag{15}$$

  solved by Cholesky on $(X_c^\top X_c + \lambda I)A = X_c^\top Y_c$ and $b = \bar{y} - \bar{x}A$, with $\rho = 10^{-4}$. Per-layer maps compose into one span map $A_{\mathrm{tot}} \leftarrow A_{\mathrm{tot}} A$, $b_{\mathrm{tot}} \leftarrow b_{\mathrm{tot}} A + b$, so the group becomes one affine block $\hat{B}(x) = A_{\mathrm{tot}} x + b_{\mathrm{tot}}$.
- `NonlinearMultiply`: a new trainable GPT-2 block, initialised from the *current* left merged block, so chains `1-2→1-3` start from the already-learned block and learn the full original span; only its parameters receive gradients.

Objectives for the learnable block.. With teacher input h and output $h^\star$, the student block maps $h \mapsto \hat{h}$; the hidden loss is a scale-normalised MSE (teacher-forced) and the distillation term at $T = 2$ is

$$\mathcal{L}_{\mathrm{hid}} = \frac{\|\hat{h} - h^\star\|_F^2 / N}{\|h^\star\|_F^2 / N + \epsilon}, \quad \mathcal{L}_{\mathrm{KL}} = T^2 \cdot \frac{1}{N} \sum_n \mathrm{KL}\big(p_n^t \,\|\, p_n^s\big), \quad \mathcal{L} = \lambda_{\mathrm{hid}} \mathcal{L}_{\mathrm{hid}} + \lambda_{\mathrm{KL}} \mathcal{L}_{\mathrm{KL}}, \tag{16}$$

Model	PPL	Top-1	Removed
Original GPT-2	29.37	0.3826	0.00%
Learned k=1	30.40	0.3763	5.70%
Learned k=5	47.31	0.3258	28.48%

Table 8: Experiment A on official WikiText-2 validation (244,824 scored tokens).

k	WikiText-2	DBpedia-14	AG News	IMDb	Removed
0	29.37	40.44	62.40	36.90	0.00%
1	30.40	40.46	66.54	38.09	5.70%
2	33.06	44.18	77.96	42.34	11.39%
3	36.49	50.55	90.15	48.02	17.09%
4	41.74	59.90	112.07	57.10	22.78%
5	47.31	70.41	139.15	68.23	28.48%

Table 9: Experiment A final evaluation: learned-model perplexity by number of merged pairs k and dataset. WikiText-2 uses its official validation split (244,824 scored tokens); the other three use one-shot test shards (32,767 scored tokens each). At k=5 the DROP-FIRST control reaches 98.41, 106.72, 173.99 and 90.52. All values are the raw learned model; a guarded deployment rule falls back to DROP-FIRST per pair when the ranking margin fails, but no selected plan triggers it.

with $\epsilon = 10^{-8}$, $\lambda_{\text{hid}} = \lambda_{\text{KL}} = 1$, and no ground-truth CE. Optimisation: AdamW, LR 3×10^{-5}, weight decay 0.01, gradient clipping 1.0, at most 15 epochs with patience 15; epoch 0 evaluates the untrained block. Data budgets: 20,480 local tokens ($\mathcal{L}_{\text{hid}}$), 20,480 distillation tokens ($\mathcal{L}_{\text{KL}}$) and 4,096 holdout tokens, all from the WikiText-2 *train* split.

Results: in-domain selection.. Table 12 reports the chosen plan and WikiText-2 validation perplexity at each stage (original 28.61). At one step `NonlinearMultiply` on blocks 0-1 reaches 29.40 and `Remove` reaches 30.01, whereas `Average` and `LinearMultiply` already cost 47.10 and 36.36. After five steps `Remove` and `NonlinearMultiply` are the only methods below 100 perplexity. Plans mix disjoint and chained merges.

Results: complete unseen-test matrix.. Table 13 gives the 5-stage×4-method unseen perplexity matrix per dataset. `NonlinearMultiply` is the best method at every stage on AG News and IMDb, and from stage 2 onward on DBpedia; the single exception is DBpedia stage 1, where `Remove` edges it by 0.03.

Structural verification.. Five self-checks run on a randomly initialised tiny GPT-2 before any compression: (i) disjoint and chained plans are reconstructed consistently; (ii) multi-step removal keeps exactly the intended first block; (iii) averaging is uniform over original blocks; (iv) a nonlinear block is initialised from the current left merged block; and (v) only the newly created block has `requires_grad`. Config, plans and checkpoints are hash-locked and the unseen test is run exactly once.

3.4 Experiment C: Affine Merging with SG-SC and CGOS

Idea.. Treat each block as an affine map, then ask three questions about it. Do the obvious fixed operators (deletion, averaging, plain composition) hold up? Does replacing the fixed rule with a similarity-gated, error-weighted SLERP version (SG-SC) help? And can a calibration-based selector (CGOS) pick the right operator per pair without extra hyperparameters?

Affine composition.. Using a row-vector convention a block is $M_\ell(x) = xA_\ell + b_\ell$. For adjacent maps (A_1, b_1) and (A_2, b_2):

$$\texttt{Remove} : (A, b) = (A_2, b_2), \quad \texttt{Average} : (A, b) = \left(\tfrac{A_1+A_2}{2}, \tfrac{b_1+b_2}{2}\right), \quad \text{Multiply} : (A, b) = (A_1A_2,\ b_1A_2 + b_2). \tag{17}$$

Random-Weight Average (RWA) permutes the entries of the second map within each tensor while preserving its distribution before the 50–50 average.

SG-SC.. Similarity-Gated, Error-Weighted SLERP Composition combines a compatibility gate (equally weighted CKA and cosine similarity of pooled hidden-state means) with calibration errors from `Average` and Multiply to set the interpolation weight, after which

$$\text{slerp}(a, b; w) = \frac{\sin((1-w)\theta)}{\sin\theta} a + \frac{\sin(w\theta)}{\sin\theta} b, \qquad \theta = \arccos\frac{a \cdot b}{\|a\|\|b\|}. \tag{18}$$

Dataset	Orig. top-1	Learned top-1	Orig. top-5	Learned top-5
WikiText-2	38.26%	32.58%	60.98%	54.19%
DBpedia-14	38.00%	30.49%	58.20%	50.10%
AG News	32.44%	22.53%	52.31%	41.44%
IMDb	34.18%	26.34%	57.83%	49.22%

Table 10: Experiment A next-token top-1 and top-5 accuracy of the original model and of the learned k=5 model on the same scored tokens as Table 9.

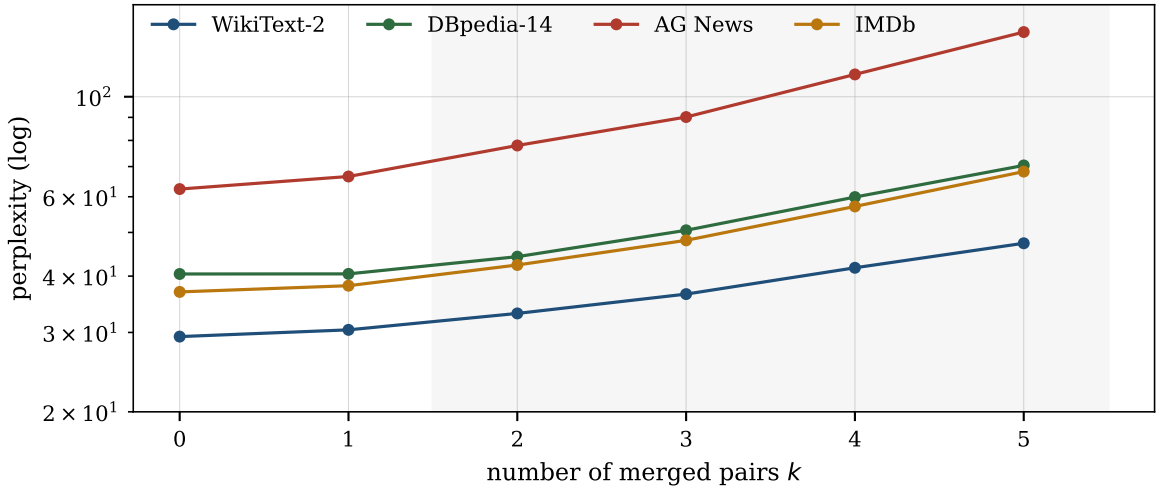


Figure 1: Experiment A: learned-model perplexity by number of merged pairs k, per dataset (log scale). The curves rise smoothly with depth, and at k=5 the learned model stays far below the DROP-FIRST control on every dataset (98.41–173.99, Table 9).

CGOS.. Calibration-Guided Operator Selection computes, per adjacent pair, the calibration hidden-state reconstruction error of `Remove`, `Average` and Multiply, and selects the smallest-error operator; it adds no hyperparameters, CKA, SLERP or interpolation.

Protocol and results.. Affine maps are fitted on a training portion; diagnostics for CGOS come from a separate calibration portion; validation perplexity ranks the candidates; one joint selection then fixes the final configuration, and the test set is used once, at the very end. The FULL configuration uses 20,000 fit tokens, 20,000 calibration tokens and 50,000 eval tokens; RWA averages five random runs; latency is measured over 30 repetitions at seed 42 in float32. The joint selection picks **Multiply for pair** 6–7, at validation PPL = 41.5146 and test PPL = 47.9306 (original test 38.4868), with 10.917% parameter reduction and compressed latency 18.51±0.41 ms versus 22.73±5.07 ms baseline; CGOS agrees with the validation-best operator on 9 of 11 pairs (81.82%).

3.5 Experiment D: Calibrated Similarity-Guided SLERP

Idea.. The idea is to let the model tell us which pair to merge, rather than guessing. Adjacent pairs are ranked by how similar their hidden states are, the chosen pair is merged with SLERP at a weight tuned on a handful of calibration tokens—separate weights for attention and MLP—and the process repeats, with no gradient training anywhere.

Method.. For every adjacent pair $(L, L{+}1)$ we compute linear CKA and the cosine similarity of their z-scored block outputs and fold them into a single score S; the top-k pairs are shortlisted. The merge initialises the SLERP weight from a similarity prior $w = g(S)$, then tunes it on the calibration set with per-group weights $w_{\text{attn}}, w_{\text{mlp}}$ and a mode search over {SLERP, LERP}. Calibration draws ~4k tokens for similarity and ~8k tokens for tuning, all from the WikiText-2 *train* split. The procedure is greedy and iterative: after each merge the similarities are recomputed on the new model.

Results.. Table 15 summarises the runs. On the calibration-selected pair the fixed-0.5 linear merge collapses (val 111.36) and affine Multiply lands at 53.32; the calibrated SLERP (OURS) reaches val = 38.30 and test = 43.85 at one merge—within a point of deletion's own val = 40.87, but with a smooth interpolant in between instead of a hard spatial cut. Pushing to four merges degrades the val to 59.17 (test 66.88), a gentler slope than either the linear

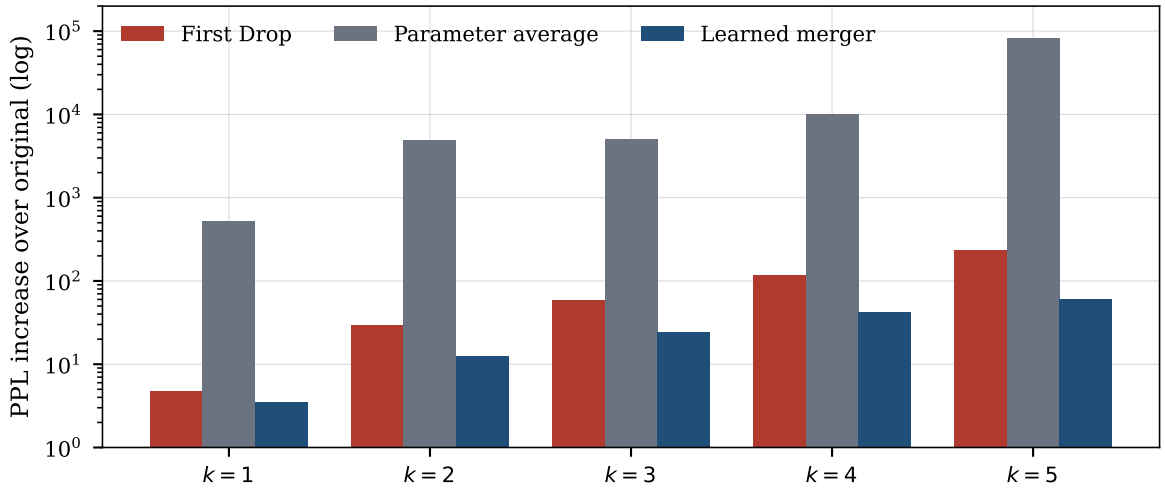


Figure 2: Experiment A: perplexity increase over the original GPT-2 on WikiText-2 validation for the three methods at k=1 . . . 5 (log scale). Parameter averaging collapses immediately; the learned merger stays below DROP-FIRST at every level.

Epoch	Holdout total loss	New best
0	0.2696	0.2696
3	0.1549	0.1549
6	0.1291	0.1291
9	0.1238	0.1238
12	0.1174	0.1174
15	0.1136	0.1136

Table 11: Experiment B ablation over nonlinear training length for the selected stage-1 span 0-1 on the train-only holdout; epoch 15 is selected. This is the only hyperparameter ablated.

or the affine route.

3.6 Foundational Exploration: Design-Space Map, Results and Limits

The first author ran a broad, *tangential* exploration that enumerated and stressed the projects and techniques on which the four primary experiments are built, producing the reference baselines and the reproducibility methodology shared by the whole group. Fifteen approaches spanning eight mechanism families (influence-based removal, low-rank/bridge replacement, quantization, weight merging, parameter tying, distillation, learned sparsity/structured pruning, and adaptive depth) were implemented and probed on two architecturally unlike models, GPT-2 small and SmolLM-135M. Four produced usable results —the fitted ternary bridge, the Block-Influence control, SVD low-rank and the quantized no-op bypass; distillation repair and LayerDrop produced results only after a budgeted re-run with per-fit seeds; the affine-bridge, permutation-averaging, weight-merging and weight-tying variants failed at increasing multiples of perplexity; and the remaining notebooks stayed scaffolds at the exploration's compute budget. The per-approach artefacts, negative results and the full results table are archived under `https://github.com/sumitjhaa/gpt2-layermerging/tree/main/Assignment2`; we quote only the numbers that survived our own audit below.

Block Influence and controlled removal.. For every decoder block $\mathrm{BI}_l = 1 - \cos(h_{\text{in}}^{(l)}, h_{\text{out}}^{(l)})$, computed from one forward pass; low BI means near-identity. The rankings are GPT-2: $4, 1, 2, 3, 5$ and SmolLM-135M: $2, 27, 5, 4, 3$ (most to least redundant). Targeted removal beats *random* removal decisively for $k \geq 2$ (see Figure 3) (GPT-2 k=2: 47.95 vs. 139.09; k=3: 121.81 vs. 4181.21; k=4: 943.24 vs. 4237.94; SmolLM k=3: 29.91 vs. 34.23; k=4: 34.20 vs. 57.37), while k=1 is cheap either way. This is the controlled evidence that BI ranking is what makes pruning cheap, and it is the null control used throughout.

Fitted ternary bridge (TERGRASP).. Replacing a redundant block by a bridge $f(x) = x + xUV$ with U, V *fitted* on captured (X, Y) pairs, $\min_{U,V} \|XUV - (Y - X)\|_F^2$ (Adam, 80 iters), then AbsMean-quantised to $\{-1, 0, +1\}\gamma$ (Ma et al., 2024), is the headline exploration result. It produces *identical* parameter counts to plain deletion (a rank-32 bridge is 49,152 weights standing in for a 7.1M block) and keeps the graph depth while shrinking the replaced

Step	Method	Chosen plan	Val PPL
0	Original	–	28.61
1	`LinearMultiply`	6-7	36.36
1	`Remove`	0-1	30.01
1	`Average`	7-8	47.10
1	`NonlinearMultiply`	0-1	**29.40**
2	`LinearMultiply`	6-8	44.79
2	`Remove`	0-1; 8-9	35.12
2	`Average`	6-8	76.03
2	`NonlinearMultiply`	0-1; 3-4	–
3	`LinearMultiply`	6-9	60.75
3	`Remove`	0-1; 2-3; 8-9	43.07
3	`Average`	5-8	138.96
3	`NonlinearMultiply`	0-1; 3-4; 7-8	–
4	`LinearMultiply`	1-2; 6-9	87.22
4	`Remove`	0-1; 2-3; 7-9	58.03
4	`Average`	4-8	238.95
4	`NonlinearMultiply`	0-1; 3-4; 5-6; 7-8	–
5	`LinearMultiply`	1-3; 6-9	122.05
5	`Remove`	0-1; 2-3; 5-6; 7-9	89.10
5	`Average`	3-8	424.98
5	`NonlinearMultiply`	0-1; 3-4; 5-6; 7-8; 9-10	–

Table 12: Experiment B chosen plan and WikiText-2 validation PPL by step. The `NonlinearMultiply` stage-1 score is 29.4023; its stage-2–5 validation numbers were not captured (marked `--`); their quality is reported on unseen data.

weights to ∼1.58 bit/param. Under per-fit seeding it is clearly worse than deletion (GPT-2 k=1: 58.67 ± 9.53 vs. 45.76; SmolLM 30.77 ± 3.54 vs. 19.93). Its value is representational rather than a perplexity win: the computation graph keeps its full depth while the replaced weights shrink to ∼1.58 bit/param. Single-seed sweeps over rank, bits and calibration text confirm the picture and add one nuance (Table 18): giving the bridge more capacity makes GPT-2 perplexity *worse*, and enlarging the fit text from 2k to 20k characters is the only axis that clearly helps—GPT-2 drops to 47.86, still above deletion's 45.76.

Repair and LayerDrop.. Pruned distillation $\mathcal{L} = \alpha\,\mathrm{CE} + (1-\alpha)T^2\mathrm{KL}$ (Ma et al., 2023) recovers a large fraction of the deletion cost at no parameter cost (healed beat pruned in 12/12 runs), and LayerDrop (Fan et al., 2020) is the strongest method on SmolLM-135M and the only configuration that lands *below* its own unpruned baseline (15.54 ± 0.01 vs. 16.39)—evidence that the best method is *model-dependent* (Table 17).

Supporting rules.. An SVD low-rank surrogate on SmolLM equals the baseline at rank 256 (19.93), gives 22.00 at 128 and 24.22 at 64. The quantized no-op bypass shows the three lowest-BI layers survive 2-bit quantization with no perplexity change (SmolLM L5: 17.76 → 17.76), so the same ranking predicts both removability and quantization tolerance. A training-free SVD diagnostic forms the residual map $\Delta = Y - X$ and shows Block Influence and residual low-rankness are *different* properties (lowest-BI GPT-2 block L4 has rank-32 explained variance 0.392 while L1 has 0.875), explaining why local affine fitting improves while end-task perplexity can degrade.

Reproducibility audit.. A global-RNG defect was found and fixed: the same configuration gave 47.67/27.04 when re-executed alone but 47.95/27.67 when reached by running the notebook top to bottom. Every stochastic fit is now driven from an explicit per-fit `torch.Generator`, and a closing audit of thirteen structural invariants refuses to certify output unless all hold.

Limitations of the foundational exploration.. *(i) Compute.* All fifteen approaches ran on a single free Apple-Silicon MPS laptop with no paid GPU and training budgets capped at 60–80 steps; consequently several training-heavy approaches (KD repair, KD student, learned L_0 gates, LayerDrop as first implemented, early exit) produced no separable results and are reported as archived scaffolds rather than findings. *(ii) Confounded bridge.* The bridge is rank-32 *and* 2-bit simultaneously and no unquantized float32 control was run, so the honest claim is narrow ("this particular linear-plus-residual form loses to deletion, and giving it more capacity did not help"), not "low-rank re-

DBpedia-14 (orig. 30.99)				
Stage	Remove	Avg.	Linear	Nonlin.
1	**32.24**	52.65	52.29	32.27
2	36.99	88.29	67.42	**35.53**
3	43.83	153.47	108.49	**40.78**
4	57.64	246.82	191.19	**50.07**
5	85.17	424.94	294.50	**59.81**

IMDb (orig. 37.20)				
Stage	Remove	Avg.	Linear	Nonlin.
1	38.47	55.02	53.00	**38.31**
2	42.76	84.65	70.58	**41.97**
3	52.20	133.21	105.11	**46.84**
4	65.77	201.24	183.09	**55.63**
5	85.12	310.88	282.00	**66.03**

AG News (orig. 51.82)				
Stage	Remove	Avg.	Linear	Nonlin.
1	53.34	88.67	96.83	**53.04**
2	62.78	146.54	131.50	**63.19**
3	78.18	249.51	211.44	**72.98**
4	104.43	381.09	394.92	**96.04**
5	150.97	569.19	613.16	**115.97**

Relative degradation (%)					
Data	St.	Rem.	Avg.	Lin.	Non.
DBp.	1	**4.0**	69.9	68.7	4.1
DBp.	5	174.8	1271.1	850.2	**93.0**
AG	1	2.9	71.1	86.9	**2.4**
AG	5	191.4	998.5	1083.3	**123.8**
IMDb	1	3.4	47.9	42.5	**3.0**
IMDb	5	128.8	735.6	658.0	**77.5**

Table 13: Experiment B unseen perplexity by dataset (top) and relative degradation versus each corpus's original (bottom). Best per row in bold.

Metric	Value	Metric	Value
Original test PPL	38.4868	Compressed parameters	110,854,656
Selected pair	6–7	Parameter reduction	10.917%
Selected method	Multiply	Compressed latency	18.51 ± 0.41 ms
Selected validation PPL	41.5146	Baseline latency	22.73 ± 5.07 ms
Final test PPL	47.9306	CGOS agreement	9/11 = 81.82%

Table 14: Experiment C FULL-run summary (GPT-2, WikiText-2, all 11 pairs).

placement loses". *(iii) Broken GPT-2 SVD row.* The GPT-2 SVD row is flat at 36.78 because a dtype guard swallowed the weight edit; it needs an explicit `.float()` rerun and is unusable as evidence. *(iv) Noisy quantization trend.* The GPT-2 quantization trend is invisible at the 3k-token evaluation budget and is eval noise. *(v) Protocol mismatch.* The exploration uses 256-token non-overlapping windows over a 3k-token budget, whereas A–D use exact sliding-window evaluation, so these numbers are *not* directly comparable to A–D and are used qualitatively. *(vi) Single-seed ablations.* The rank and bitwidth sweeps are single-seed and should be read as indicative. *(vii) Superseded numbers.* The early single-seed bridge value (45.11 at k=1) came from the global-RNG configuration and was replaced by the per-fit-seeded runs. These limits are the reason the exploration is presented as the *map* of the design space rather than as a competing headline method.

3.7 Additional Pilot: Closed-Form Training-Free Merges

A closed-form, training-free pilot explored three further mechanisms on GPT-2 (quick-mode: 8192 fit tokens, one seed; the full 50k-token/three-seed grid is pending). *Remove-and-Patch (R&P)* keeps one real block and replaces the other by a mean plus a fitted low-rank correction $y = x + \mu + (x - \bar{x})D_r$ with D_r from the top-r right singular vectors of $X_c D_{\text{ols}}$ (Eckart–Young), stored factored as (U, V). *Union-Select-Reconstruct (USR)* scores all 6144 MLP neurons of both blocks by $s = \mathbb{E}[a^2]\|o\|^2$, keeps the top 3072, and re-solves the readout with shrinkage. *Correlation-Guided Neuron Fusion* pairs neurons by Pearson correlation, averages their inputs and jointly refits outputs. In the quick run R&P reaches the best per-pair validation values (e.g. pair 2–3: 32.44), random-subspace and random-pairing controls are uniformly worse (validating the learned structure), and pair-local normalised MSE predicts end-to-end damage (Spearman 0.814); early-pair (0–1) merges are catastrophic for every method. These numbers are preliminary.

3.8 Cross-Experiment Analysis and Merged Results

Because the four experiments used different but related exact evaluators (different window sizes, strides and token caps), absolute perplexities are not directly comparable across experiments; we therefore compare methods *within* each experiment and only look at orderings that repeat. Three patterns came up over and over across the portfolio.

Method	Merges	Params	Val PPL	Test PPL
Original GPT-2	0	124,439,808	33.94	38.83
A1: remove layer 5	1	117,351,936	40.87	46.69
A1: linear average 0.5	1	117,351,936	111.36	120.59
A1: affine multiply	1	110,854,656	53.32	57.47
OURS (calibrated SLERP)	1	117,351,936	**38.30**	**43.85**
OURS	2	110,264,064	41.80	47.25
OURS	3	103,176,192	48.66	55.27
OURS	4	96,088,320	59.17	66.88

Table 15: Experiment D summary. The selected pair is 3–4 with similarity $S = 0.951$ and a tuned weight $w = 1.0$.

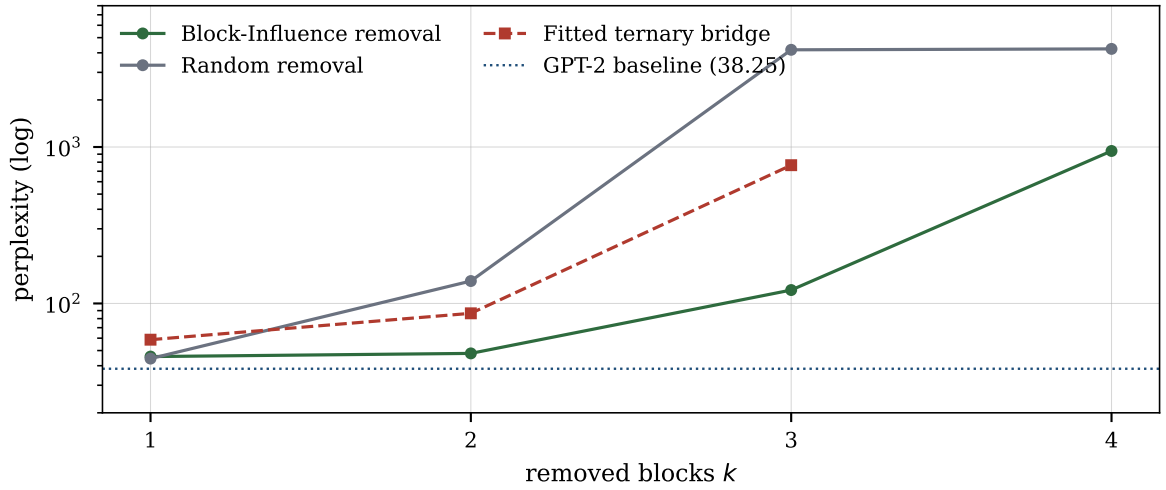


Figure 3: Foundational exploration on GPT-2: perplexity of Block-Influence removal, random removal and the fitted ternary bridge at k=1 . . . 4 against the 38.25 baseline (log scale). BI removal is the only rule that survives beyond k=2; the bridge loses to deletion at identical parameter counts.

(1) Learned fusion dominates at depth.. In Experiment B `NonlinearMultiply` is best in 14 of 15 unseen stage×dataset cells; in Experiment A the guarded learned merger beats DROP-FIRST at every k on the ranking holdout (CE gap −0.025 to −0.805 nats) and on IMDb by 68.23 vs. 90.52 at k=5. Training-free operators fall behind increasingly with depth.

(2) Averaging is unusable.. `Average` reaches +1271.1% on DBpedia at stage 5 (B), val = 111.36 at a single merge (D), and 10^2–$10^3\times$ baseline in the foundational sweep. The mechanism is the neuron-permutation mismatch between adjacent blocks, which the learned/calibrated operators explicitly repair.

(3) The best training-free rule is model- and pair-dependent.. Affine Multiply is best on pair 6–7 under CGOS (C); calibrated SLERP is best at one merge under similarity selection (D); in the foundational study LayerDrop wins on SmolLM while distillation wins on GPT-2. No single universal rule is supported.

3.9 Discussion and Limitations

Threats to validity.. First, the experiments use different exact evaluators, so absolute perplexities are not mutually comparable and only within-experiment orderings are claimed. Second, Experiment A adapts a merger to one GPT-2 checkpoint; its accuracy is next-*token* BPE accuracy, not whole-word; its ten passes mix single and joint tasks; and its additive-cost selection is a surrogate that can miss a better joint plan. Third, Experiment B's selection is greedy and stage-wise and its `NonlinearMultiply` stage-2–5 validation numbers were not captured, so those in-domain curves are incomplete rather than favourable. Fourth, Experiment C evaluates one merge depth and one selected pair and does not sweep the affine ridge; Experiment D has no unseen-domain evaluation in the reported runs and its results come from notebook artefacts. Fifth, the pilot's numbers are quick-mode and single-seed, and the foundational exploration's limits are itemised in Section 3.6. Sixth, the guard in A and the hash lock in B are held-out *selection* rules, not guarantees on unseen domains. Seventh, we did not re-run FuseGPT, which has no official release, so that

Model	Operator	k=1	k=2	k=3	k=4
GPT-2	baseline		38.25		
GPT-2	BI removal	45.76	47.95	121.81	943.24
GPT-2	random removal	44.42	139.09	4181.21	4237.94
GPT-2	fitted ternary bridge	58.67	86.49	765.21	—
SmolLM	baseline		16.39		
SmolLM	BI removal	19.93	26.13	29.91	34.20
SmolLM	random removal	20.83	26.37	34.23	57.37
SmolLM	fitted ternary bridge	30.77	41.77	58.21	—

Table 16: Foundational exploration: the fitted ternary bridge loses to deletion at identical parameter counts, while the BI ranking beats random removal from $k \geq 2$ (all GPT-2 bridge/random values are 3-seed; the earlier single-seed TerGRASP value 45.11 came from an unreproducible global-RNG run and is superseded).

Model	Method	k=1	k=2	Recovered
GPT-2	BI removal (pruned)	45.76	47.95	—
GPT-2	BI + distillation	**41.68**	**44.06**	54.3%
GPT-2	LayerDrop	51.74	57.53	—
SmolLM	BI removal (pruned)	19.93	26.13	—
SmolLM	BI + distillation	18.67	22.14	34.9%
SmolLM	LayerDrop	**15.54**	**19.62**	beats baseline

Table 17: Foundational exploration: repair and LayerDrop (validation perplexity, mean over three seeds). LayerDrop alone beats its own 16.39 baseline on SmolLM; distillation wins on GPT-2.

comparison is qualitative. Finally, all experiments share one backbone: no transfer to other checkpoints is claimed.

Cost.. Learnable methods (A, B) strictly cost more than training-free ones; their advantage is a quality result, not a compute result. A's 39.4M-parameter merger is discarded at deployment. `LinearMultiply` (B) is the most expensive training-free operator because it needs activation collection and a Cholesky solve per span per stage.

Multilingual extension.. The mentor suggested applying the same merges to multilingual models. Every operator here is backbone-agnostic, and a multilingual checkpoint would test whether the fusion advantage survives richer token statistics and faster LayerNorm/attention drift. We leave this to future work.

Takeaways.. Putting the four experiments together, the story is consistent. Once compression goes past one or two merges, something trained is doing the work: the learnable methods (A, B) are the only ones that hold up at depth, and their lead over training-free operators grows as more blocks are removed. Averaging two blocks is not a merging strategy—the neurons simply do not line up—except when an entire contiguous tail span is averaged, and even then it is still beaten. Among training-free rules the winner depends on the model and the pair: affine composition on one pair (C), calibrated SLERP on another (D), LayerDrop on small models. That is why we report every per-pair ranking rather than a single recipe. If we had to pick one cheap baseline to reuse, it would be Block-Influence removal patched with distillation.

3.10 Code and Reproducibility

Every method in this report was implemented from scratch by the group; no external compression library was used. The tangential exploration and its full results table (fifteen approaches, per-fit seeds, CSV dumps), the four experiments, the pilot and the source of this document are all compiled in one place: `https://github.com/sumitjhaa/gpt2-layermerging/tree/main/Assignment2`. Each experiment ships as a notebook with saved CSV artefacts, so every number in Table 19 traces back to a raw run file rather than a re-derivation.

- **Tooling.** Standard Python/PyTorch with HuggingFace `transformers` and `datasets`; the dependency list sits beside each notebook. Pretrained weights (`openai-community/gpt2`, `HuggingFaceTB/SmolLM-135M`) and WikiText-2 are downloaded and cached once.
- **Compute.** Experiment A ran on two NVIDIA T4 GPUs with one frozen teacher and one merger replica per GPU; the other experiments and the exploration ran on a single GPU or an Apple-Silicon laptop under the budgets stated in their sections.

Sweep	Value	GPT-2 PPL	SmolLM PPL
Bridge rank	4	50.57	26.54
	16	60.35	25.22
	64	68.18	26.02
	128	341.10	32.83
Quant. bits	2	52.35	34.83
	4	69.78	44.02
	8	69.72	39.55
Calib. text	2k chars	67.45	32.92
	20k chars	47.86	24.12

Table 18: Foundational exploration, single-seed sweeps on the fitted bridge at k=1: bridge rank, AbsMean quantisation bits, and calibration text length (one axis changes at a time). Deletion sits at 45.76 (GPT-2) and 19.93 (SmolLM). More bridge capacity makes GPT-2 perplexity *worse*, while the larger calibration text is the only axis that clearly helps.

Exp.	Family	Best artifact	Selection signal	Key result
A	Learnable, weight-space	Hypernetwork correction	Guarded ranking	IMDb 68.23 vs. 90.52 at k=5 (28.48% removed)
B	Learnable, activation-space	Nonlinear block	Exact validation NLL	Best in 14/15 unseen cells; IMDb +77.5% vs. deletion +128.8%
C	Affine + selection	Multiply on 6–7 (CGOS)	Calib. recon. error	Test 47.93 at 10.92% reduction; CGOS agrees 9/11
D	Interpolation	Calibrated SLERP	Similarity + calibration	Val 38.30 / test 43.85 at one merge
Found.	Reference baselines	BI removal + distillation	Validation NLL	Recovers 54.3% of deletion cost on GPT-2
Pilot	Closed-form	Remove-and-Patch	Validation PPL	Best per-pair values; controls worse (quick run)

Table 19: Headline configuration of each study on GPT-2. Absolute numbers are each experiment's own protocol and are not mutually comparable; the ordering of methods within an experiment is the reliable signal.

- **Determinism.** Every stochastic fit in the exploration is driven by an explicit per-fit `torch.Generator` seed since the global-RNG audit (Section 3.6); Experiment B locks its protocol with SHA-256 hashes; Experiment A asserts its six preflight checks before any training starts.
- **Data handling.** The DBpedia-14, AG News and IMDb shards are opened exactly once for final reporting, with the scored token positions recorded, so the unseen numbers can be re-checked without re-selecting.

4 Team Contributions

The six members worked as one team on adjacent-layer merging. Following the assignment's instruction to assign contributions fairly, and in recognition that each member owned one mechanistically distinct experiment *and* shared the common protocol, we record an **equal contribution** for all six members (100%/6 ≈ 16.67% each). Table 20 lists the members with their roll numbers and primary deliverables; the paragraphs below give the auditable detail.

Member	Roll No.	Primary deliverable in this report	Share
Sumit Kumar Jha	26210180	Tangential exploration and reference baselines	16.67%
Amrita Lodh	26290001	Exp. A: learnable weight-space merging	
Hiroyoshi Tsujimoto	26210172	Exp. B: four-operator iterative comparison	
Sura Sravan Kumar	24110362	Exp. D: calibrated SLERP merging	
Satvik Sharma	26120008	Pilot: closed-form training-free merges	
Lingala Hasini	24110186	Exp. C: affine merging, SG-SC and CGOS	

Table 20: Team contribution summary. All six members contributed equally (100%/6 ≈ 16.67% each); the table lists each member's roll number and primary deliverable in this unified report.

- **Sumit Kumar Jha (equal, 16.67%).** Tangential exploration and design-space mapping: the fifteen-approach survey across eight families on GPT-2 and SmolLM-135M, the reference baselines (Block-Influence removal, distillation repair, LayerDrop), the fitted ternary bridge, the reproducibility and seed audit, the shared repository and evaluation protocol, and the integration and LaTeX production of this merged report.

- **Amrita Lodh (equal,** 16.67%**).** Experiment A: the second-block-anchored learnable weight-space merger (novelty, architecture and weight tokenisation, training objective, two-GPU synchronous schedule and batch auto-tuner, pair ranking and guarded selection), all Experiment A experiments and analysis, and the Experiment A report.
- **Hiroyoshi Tsujimoto (equal,** 16.67%**).** Experiment B: the four-operator suite (removal, averaging, closed-form affine linear composition, and a trainable block), the exact sliding-window evaluator, the structural self-checks and hash-locked protocol, greedy stage-wise selection, the full three-dataset, five-stage, four-method unseen matrix, and the Experiment B report.
- **Sura Sravan Kumar (equal,** 16.67%**).** Experiment D: the calibrated, similarity-guided SLERP merger (CKA+cosine pair scoring, similarity-prior and calibrated per-group attention/MLP weights, mode search, greedy iteration), the comparison against MKA/CoM-style baselines, and the Experiment D report.
- **Satvik Sharma (equal,** 16.67%**).** The closed-form training-free pilot: Union-Select-Reconstruct (USR), Remove-and-Patch (R&P) and Correlation-Guided Neuron Fusion, with random-subspace and random-pairing controls, the pair-local nMSE predictor of end-to-end damage, and the pilot report.
- **Lingala Hasini (equal,** 16.67%**).** Experiment C: affine merging, the Random-Weight-Average control, SG-SC and its ablations, CGOS calibration-guided operator selection, the FULL-run evaluation with latency measurement, and the Experiment C report.

Shared.. Dataset protocol (WikiText-2 for training/selection; DBpedia-14, AG News and IMDb for one-shot unseen testing), the common baseline/SOTA framing, weekly mentor meetings, and cross-checking of each other's numbers against the raw artefacts. Because the work of every member is included in this single report, the group submits one unified document and all members receive the same credit.

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